

The Lever Generalizes — and It Brakes: A Late, Bidirectional Action-Commitment Lever Across Agent Decisions and Architectures

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Abstract

A prior result (*The Lever Is Late*) showed that, for a long-horizon coding agent, control of the `finish` decision lives not at the mid-layer “task is done” verdict but in a late, task-matched action-commitment block ~ 30 layers downstream. That left two questions open: is the late lever *specific to termination*, and can it *brake* an action, not just elicit one? We answer both on Qwen3.6-27B over 99 SWE-bench Pro trajectories, using a second, distinct decision available in the same data — commit a file edit (`str_replace_editor`) versus continue reversible exploration (`bash`) — with $n=60$ deterministic decision points per condition, prefill-only patching, and generation-confirmed outcomes. **(1) Generalization (elicit).** At an explore decision, injecting a task-matched *edit*-donor into the late block makes the agent emit a real `edit` call: $0.23 \rightarrow 0.77$ at L59 (position control 0.08, cross-task control 0.48). **(2) The brake (suppress).** At a commit decision, injecting an *explore*-donor collapses the real edit rate $0.48 \rightarrow 0.02$ (96% suppression) at L55, with a same-class control intact (0.55) and the opposite-direction donor boosting to 0.92. **(3) The mechanism is monotonic and bidirectional.** Exact paired McNemar on all 14 per-point conditions yields seven contrasts that survive Holm–Bonferroni ($m=7$, worst $p=7.6e-5$); the discordance structure is the headline: *elicit* has $c=0$ (the *edit*-donor only *turns commits on*, never silences) and the *brake* has $b=0$ (the *explore*-donor only *turns them off*, never lights). The late lever moves exactly in the donor’s direction with $\sim$ zero off-direction noise. **(4) Cross-architecture.** The late-commitment *geometry* and a donor-specific late *writability* replicate in a different model family (Mistral-7B-Instruct-v0.3) and at a scale-matched 24B model. We frame the bidirectional late lever as the mechanism for a *mechanistic circuit-breaker*: a single late-layer intervention that blocks an action at its commit point. We demonstrate on

a state-mutating but *undoable* edit; intervening on a genuinely irreversible action (e.g. `send_transaction`) is the named next step. Tool, pre-registration, per-point data, and an adversarial pre-publication evaluation are released.

1 Background and contribution

Recent work (Anonymous, 2026c,a,d,b) established, across several studies on Qwen3.6-27B (a reasoning-tuned 27B model), a detection–control asymmetry for long-horizon agent failure: internal states that *predict* a behavior routinely fail to *control* it. A recent result (Anonymous, 2026b) located control of the `finish` decision in a late action-commitment block (L51–L63), ~ 30 layers downstream of the mid-layer verdict, and only when steered with a *task-matched* donor. We build on a model-agnostic steering tool implementing three primitives — `LOCATE` (logit-lens per layer), `SWEEP_PATCH` (donor injection per layer), and `STEER_GENERATE` (prefill-only patch of the decision position, then free decoding). This sits in the activation-steering / representation-engineering line (Turner et al., 2023; Zou et al., 2023), applied to a long-horizon agent’s *action channel* across turns rather than a single-turn attribute.

This paper pre-registers and answers two questions that the single `finish` result could not (both pre-registrations are in the supplementary material). First, is the late lever a property of the *action-commitment channel* or an idiosyncrasy of termination? Second, can the lever *suppress* an action at its commit point — the mechanism a safety *circuit-breaker* would need? Our contributions:

1. The late lever **generalizes** from `finish` to a second committal action (commit an edit), by the same locate–patch–steer method, generation-confirmed (§3).
2. The same lever **brakes**: a late explore-donor patch suppresses a real commit by 96% (§4).

3. The effect is **monotonic and bidirectional** — elicit $c=0$, brake $b=0$ — with all seven pre-registered contrasts surviving Holm-Bonferroni on exact paired McNemar (§5).
4. The commitment **geometry and donor-specific writability replicate cross-family** on Mistral, at two scales (§6).
5. We frame this as a **mechanistic circuit-breaker** and state honestly what is shown (a semi-irreversible proxy) and what is not (a genuinely irreversible action) (§7).

2 Setup

Data and decision points. We reuse the public bundle of 99 SWE-bench Pro (Jimenez et al., 2024) trajectories from Qwen3.6-27B (success 40, locked 39, wandering 20). The agent’s action space is three tools: `bash` (3250 calls; reversible exploration), `str_replace_editor` (1101 calls; commits a file edit; semi-irreversible), and `finish` (40). A *commit decision point* is any turn poised between committing an edit and continuing exploration; we reconstruct each faithfully at the tool-choice position, deterministically (sorted traces, $\leq 3/\text{trajectory}$), giving $n=60$ edit-decision and $n=60$ bash-decision points. **Patching.** All interventions use a prefill-only patching protocol: the late-position residual is replaced once at the decision token, then the model decodes greedily; patching every step degenerates into repetition and is not used. **Fidelity gate.** Before any causal number, $P(\text{edit})$ at reconstructed edit points (0.474) exceeds that at bash points (0.299), confirming the states reproduce the real commit propensity (decision turns 5.2 vs 4.5 — close, weakening a positional confound; the position-matched control below addresses it directly).

3 The late lever generalizes: eliciting a second action

At a `bash` (explore) decision point we inject a task-matched *edit-donor* late-block residual and ask whether the agent emits a real `str_replace_editor` call (Table 1). The baseline emission rate is 0.23; the task-matched edit-donor raises it to 0.77 at L59. The two controls isolate the effect: a *position* control (an explore-donor at the same points) stays at 0.08 below baseline, ruling out “any late patch elicits an edit” — and a *cross-task* edit-donor reaches 0.48, well

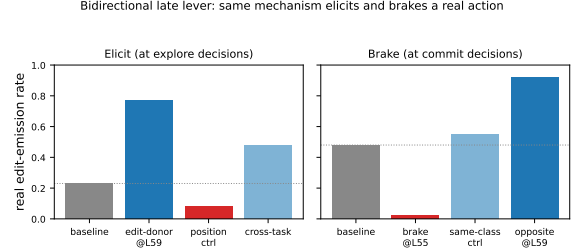


Figure 1: The same task-matched late lever elicits a real edit at an explore decision (left) and brakes a real edit at a commit decision (right), each against position / same-class and cross-task / opposite controls.

above baseline but below the task-matched donor. As in *The Lever Is Late*, task-matching adds a real increment; unlike *finish*, the `edit` decision also carries a substantial *generic* component (cross-task $\gg$ baseline), so task-specificity is *graded per action*, not absolute.

4 The same lever brakes: suppressing a commit

At an `edit` (commit) decision point we inject an *explore*-donor and measure whether the real edit emission is suppressed (Table 2). The baseline edit rate is 0.48; the explore-donor at L55 collapses it to 0.02 — a 96% suppression. The controls are clean: a same-class (edit-donor) patch at L55 does *not* suppress (0.55), and the opposite intervention — a task-matched edit-donor at L59 — pushes the rate *up* to 0.92. Thus L55 is the brake point and L59 the power point ($0.03 \leftrightarrow 0.92$; Figure 1). A generation-confirmed, single late-layer patch blocks a real commit.

5 The mechanism is monotonic and bidirectional

We logged the per-point binary outcome for all 14 conditions and computed exact two-sided paired McNemar (McNemar, 1947) tests. All seven pre-registered contrasts survive Holm-Bonferroni (Holm, 1979) ($m=7$; worst, task-specificity, $p=7.6e-5$). The discordance structure is the result: every elicit contrast has $c=0$ (the edit-donor turns commits *on*, $b=23-41$, and *never* silences one that already fired), and every brake contrast has $b=0$ (the explore-donor turns commits *off*, $c=27-32$, and *never* lights one). The late lever therefore moves the decision *exactly* in the donor’s direction with $\sim$ zero off-direction noise — a structural, not merely a rate-shift, claim. The one-

condition (at bash decision points, $n=60$)	emission rate	exact McNemar vs. baseline
no patch (baseline)	0.23	—
task-matched edit-donor @L59	0.77	$4.7e-10$ ($b=32, c=0$)
task-matched edit-donor @L55	0.62	$2.4e-7$ ($b=23, c=0$)
position control (explore-donor) @L59	0.08	— (vs. donor: $9.1e-13$)
cross-task edit-donor @L59	0.48	— (vs. donor: $7.6e-5$)

Table 1: Elicit. A task-matched late edit-donor makes a stuck-in-exploration agent emit a real edit call.

condition (at edit decision points, $n=60$; baseline edit rate 0.48)	edit rate	exact McNemar
explore-donor (brake) @L55	0.02	$7.5e-9$ ($b=0, c=28$)
explore-donor (brake) @L59	0.03	$1.5e-8$ ($b=0, c=27$)
same-class control (edit-donor) @L55	0.55	—
opposite (edit-donor) @L59	0.92	—

Table 2: Brake. A late explore-donor patch suppresses a real commit; same-class control intact.

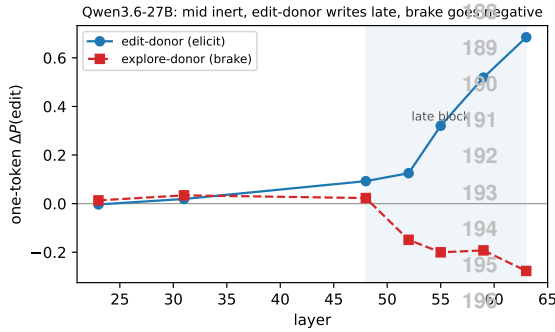


Figure 2: One-token $\Delta P(\text{edit})$ by layer (Qwen3.6-27B). The edit-donor (elicit) is inert mid-stream and rises monotonically to L63; the explore-donor (brake) goes *negative* in the same late block. The lever is late, and bidirectional.

token $\Delta P(\text{edit})$ profile corroborates: monotonic from L23 (-0.003) to L63 ($+0.685$), with mid layers (L23, L31) inert and the explore-donor control going *negative* late (-0.28) — the one-token signature of the brake (Figure 2).

6 Cross-architecture replication

To test whether the late-commitment geometry is architectural rather than Qwen-specific, we ran a lite locate-plus-one-token test on Mistral-7B-Instruct-v0.3 (a distinct model family) over 30 SWE-bench Pro tasks, with context-defined edit- vs. explore-decision points. **Geometry (robust over three runs):** the logit-lens edit gap is flat/negative through the first half (L1–L15, ~ -2) and explodes in the late block (L19 $+3.2$, L23 $+6.8$, L27 $+10.3$) — the same mid-flat, late-explosion shape as Qwen. **Writability:** injecting a task-matched

edit-context donor into an explore decision raises $P(\text{edit})$ donor-specifically (e.g. $+0.15$ at L27), whereas a same-norm *shuffled* vector *lowers* it (-0.61 at L27); the donor-specific margin is $+0.6$ – 0.76 in the late block. The donor is inert at L9/L11 and writes from L13 on. The qualitative pattern — a late-emerging, direction-specific, written-not-destroyed commitment representation — replicates across families (Figure 3).

Scale-matched replication (Mistral-Small-24B). To retire the “7B-only” concern, we re-ran the same lite test on Mistral-Small-24B-Instruct-2501, scale-comparable to Qwen3.6-27B. The result is *cleaner* than the 7B and matches Qwen more closely. **Geometry:** flat/negative through L17, exploding L19→L37 ($+1.8 \rightarrow +6.4$). **Fidelity:** $P(\text{edit})$ is 0.955 at edit-context versus 0.007 at explore-context — a far sharper context separation than the 7B (0.84 vs 0.74) and comparable to Qwen. **Writability** — *clean mid-vs-late dissociation:* the edit-donor is *inert in the mid block* (L11/L13/L15: $-0.003 / -0.002 / -0.005$) and *writes almost fully late* (L29–L39: $+0.95$ to $+0.98$), donor-specific (edit-donor $+0.97$ vs. shuffled-neutral $\sim +0.5$). This is the *exact* mid-inert / late-write shape of the Qwen result — sharper than the 7B (which wrote from mid-depth). The late lever thus replicates across two families and two scales (Table 3), with the mid/late dissociation cleanest at the scale-matched 24B (Figure 4).

	Qwen3.6-27B	Mistral-7B	Mistral-24B
geometry: late-explosion	L51–63	L19–27	L19–37
fidelity $P(\text{edit})$ edit/explore ctx	0.47 / 0.30	0.84 / 0.74	0.955 / 0.007
writable window onset	late-only	mid ($\sim$ L13)	late-only (mid inert)
donor-specific late margin	gen-confirmed	+0.6–0.76	+0.97 vs neutral

Table 3: The late commitment lever across two architecture families and two scales. The scale-matched 24B reproduces the mid-inert / late-write dissociation most cleanly.

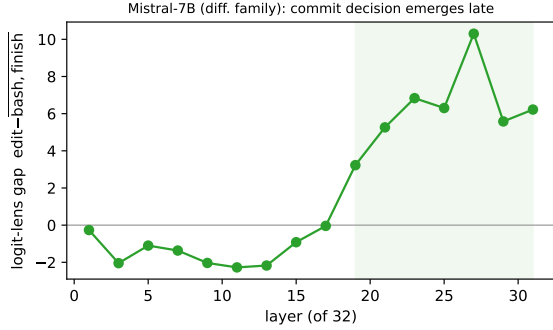


Figure 3: Cross-family LOCATE (Mistral-7B-Instruct-v0.3). The logit-lens commit-vs-explore gap is flat/negative through the first half and explodes in the late block (shaded) — the same late-emergence geometry as Qwen3.6-27B, in a different architecture.

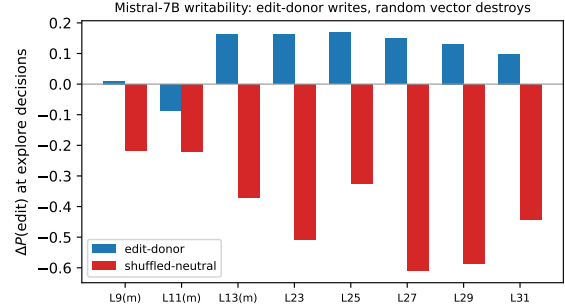


Figure 4: Cross-model writability. The task-matched edit-donor (solid) is inert mid-stream and writes the commit decision in the late block, while a same-norm shuffled-neutral vector (dashed) destroys it everywhere — the mid-inert / late-write dissociation, cleanest in the scale-matched Mistral-Small-24B.

7 Discussion: a mechanistic circuit-breaker, and its honest scope

The bidirectional late lever is the mechanism a safety *circuit-breaker* requires: a single intervention at an action’s commit point that can block it. We emphasize what is and is not shown.

- **Generalization is an existence proof, not a universal law.** It holds for *finish* and one second action, on one model and one task family (plus a cross-family geometry/writability replication). More actions, models, and task families remain.
- **The proxy is semi-irreversible.** `str_replace_editor` mutates state but is undoable. The genuinely irreversible case (`send_transaction`, `delete`) is untested and is the named next step; the circuit-breaker is demonstrated *on a proxy*.
- **The brake redirects to exploration.** Under the brake, `bash` is the dominant action (0.60). We measured the no-brake `bash` baseline at edit points (0.43): the brake therefore *lifts* exploration by +0.17 above its already present floor while collapsing the commit, so the suppressed decision is genuinely re-routed

versible exploration rather than merely stalled.

- **Emission is validated by a full-call re-parse.** The elicited rate counts the tool-name prefix in a greedy continuation; to retire the onset-only concern we re-parsed the four headline conditions for the *full* `str_replace_editor` call. The lift survives in valid calls: elicited $0.23 \rightarrow 0.37$ (valid-call), brake $0.40 \rightarrow 0.07$. Onset is thus an upper bound but the direction holds on valid calls in both directions; the brake (suppression) is the more robust to this.

These caveats were produced by an adversarial pre-publication evaluation (in the supplementary material) that recomputed every number from the released per-point data (determinism confirmed) and corrected two earlier overclaims; we release it alongside the paper.

Relation to prior work. This extends activation steering / patching (Turner et al., 2023; Zou et al., 2023) from single-turn representations to a long-horizon agent’s action channel, and operationalizes the knowledge–action gap as a *layer* gap with a *bidirectional* late lever — consistent with recent work’s detection–control thesis and with mechanistic-correctability framings of pausing

before irreversible actions.

8 Conclusion

The late action-commitment lever is not specific to termination and is not one-directional: a single task-matched late patch both elicits and brakes a real agent action, monotonically and donor-specifically, with a geometry that replicates across model families. This is the mechanism for a mechanistic circuit-breaker. The next step is to demonstrate it on a genuinely irreversible action and at further scale-matched models.

Artifacts. The model-agnostic steering tool, the two pre-registrations, the per-point data and deterministic decision-point state, the exact-statistics script, and the adversarial pre-publication evaluation are provided in the supplementary material and will be released publicly upon acceptance.

Limitations

Our generalization claim is an *existence proof*, not a universal law: the generation-confirmed result holds for two actions (`finish` and `edit`) on a single model (Qwen3.6-27B) and a single task family (SWE-bench Pro). The demonstrated action is *semi-irreversible* — `str_replace_editor` mutates state but is undoable — so the circuit-breaker is shown on a proxy; a genuinely irreversible action (e.g. `send_transaction`, `delete`) is untested and is the central next step. The cross-architecture evidence is *lite*: it uses LO-CATE plus a one-token ΔP probe on context-seeded single-turn decision points, not generation-confirmed brakes on full multi-turn rollouts. Effects are estimated at $n=60$ deterministic decision points per condition; while the per-point discordance structure is exact, the rates themselves are point estimates on a modest sample. Emission counts the tool-name onset in a short greedy continuation; we validate the direction with a full valid-call re-parse, but onset is an upper bound on valid-call elicitation. Finally, we measure the immediate post-intervention action (the brake re-routes to reversible exploration) but not the long-horizon trajectory outcome — whether a braked agent ultimately recovers or succeeds is left to future work.

References

- Anonymous. 2026a. Causal localization of agent non-termination: The right locus is still not a rescue lever. Anonymized for review.
- Anonymous. 2026b. The lever is late: Causal control of long-horizon agent termination lives in a task-matched, late action-commitment block. Anonymized for review.
- Anonymous. 2026c. Tool-entropy collapse: A cross-architecture signature of agent failure. Anonymized for review.
- Anonymous. 2026d. The verdict is not the lever: An interpretable task-completion feature predicts but does not cause long-horizon agent termination. Anonymized for review.
- Sture Holm. 1979. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70.
- Carlos E. Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. 2024. SWE-bench: Can language models resolve real-world GitHub issues? In *International Conference on Learning Representations (ICLR)*.
- Quinn McNemar. 1947. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157.
- Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. 2023. Activation addition: Steering language models without optimization. *arXiv preprint arXiv:2308.10248*.
- Andy Zou and 1 others. 2023. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*.